



Module 09

Anomaly Detection Fundamentals for OT Monitoring

Course: Cybersecurity for Embedded & OT Systems

Theme: Detect deviations early, support safety and availability

Recap — Module 08: Authentication, Authorization & Key Management in OT

- Identity management in OT is different from IT due to legacy and always-on systems
- Shared credentials create accountability and insider-risk issues
- Service accounts must follow least-privilege principles
- Credential rotation must align with maintenance windows
- Logging and audit trails provide visibility and accountability
- Authentication verifies identity; Authorization controls permitted actions
- Role separation reduces the chance of unsafe or excessive access
- Keys and secrets protect access and system trust
- Joiner–Mover–Leaver (JML) model manages access lifecycle
- Vendor access governance must be time-bound, monitored, and revoked

Bridge to Next Module:

Strong identity and access controls reduce risk by ensuring only the right users perform the right actions at the right time.

What Anomaly Detection Means in OT

In operational technology environments, an **anomaly** represents any deviation from established normal OT behavior patterns. Unlike IT security systems that primarily focus on threat detection, OT anomaly detection serves a broader operational purpose.

Anomalies can indicate various conditions requiring attention:

- Equipment faults or degradation
- System misconfigurations
- Early indicators of cyber incidents
- Operational issues affecting performance



Critical OT Principle

The primary goal in OT is **process stability and safety**, not simply "catching hackers." Anomaly detection systems must align with operational priorities.

Remember: Anomaly detection supports human decision-making. It does not replace engineering judgment, operational expertise, or safety protocols.

Why Anomaly Detection Matters in OT

OT systems operate with **predictable and repetitive** patterns by design. This characteristic makes deviations particularly meaningful and actionable. When properly implemented, anomaly detection delivers substantial operational benefits:



Early Warning

Identify potential issues before they escalate into major incidents or safety events



Faster Identification

Reduce diagnostic time for faults and operational irregularities



Preventive Maintenance

Support proactive maintenance strategies based on actual system behavior



Situational Awareness

Improve operator understanding of current system state and trends



Reduced Downtime

Minimize unplanned shutdowns and accelerate recovery efforts

Think: "A small signal detected now prevents a big shutdown later."

OT Data Sources Used for Detection

Operational technology environments generate diverse data streams that can be analyzed for anomaly detection. Understanding what data you have available is the essential first step in building effective monitoring capabilities.



Events

Discrete occurrences captured by the system: alarms, state changes, mode transitions, equipment starts and stops



Logs

Recorded actions and messages: user logins, configuration changes, command execution, system messages



Time-Series Data

Continuous measurements over time: pressure, temperature, flow rates, vibration, power consumption



Configuration Data

System parameters and settings: setpoints, thresholds, control logic, device configurations



Network Data

Communication patterns: message counts, device interactions, protocol behaviors, traffic volumes



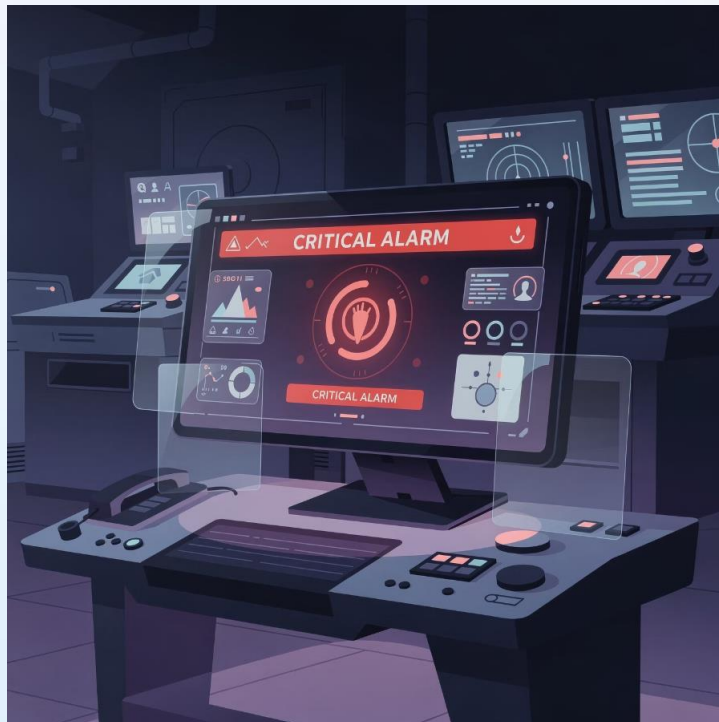
Foundation Principle: Good detection begins with knowing what data you actually have access to, its quality, and its reliability.

Events vs Logs vs Time-Series

Event

Definition: "Something happened once"

Example: High pressure alarm triggered at 14:23:07



Log Entry

Definition: "A record of an action"

Example: User 'engineer_42' logged in from workstation WS-105



Time-Series

Definition: "Value changing over time"

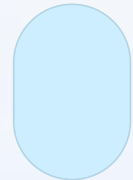
Example: Tank pressure trending from 45 PSI to 78 PSI over 30 minutes



Key Insight: Different data types reveal different anomaly indicators. Events show discrete occurrences, logs capture actions and decisions, while time-series data exposes gradual changes and trending behaviors. Effective anomaly detection leverages all three types appropriately.

Feature Extraction: What We Measure

Feature extraction is the process of selecting **simple, measurable indicators** from raw operational data. Rather than analyzing every data point, we identify specific characteristics that reveal system behavior patterns.



Messages per Minute

Communication frequency between devices



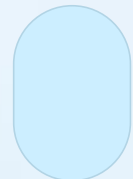
Unique Devices

Count of distinct devices active in a time window



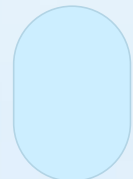
Alarm Frequency

Rate of alarm generation across the system



Error Count

Number of errors or exceptions recorded



Login Attempts

Authentication activity patterns over time



OT-Specific Requirement

In operational environments, features should be **simple, explainable, and easy to validate** with operations staff. Complex or opaque features reduce operator trust and make troubleshooting difficult. Always prioritize clarity over sophistication.

Rate-Based & Frequency-Based Indicators

Rate-Based Indicators

Concept: "How many events occur per unit of time"

Common Examples:

- Alarms per hour
- Network packets per minute
- Write operations per second
- Commands issued per shift

Frequency-Based Indicators

Concept: "How often specific patterns occur"

Common Examples:

- Polling command appears every control cycle
- Backup routine runs every 24 hours
- Status update broadcast frequency
- Maintenance window patterns

Why These Indicators Are Valuable in OT

Normal operation in OT environments exhibits **stable rates and predictable frequencies**. Production schedules, control loops, and automated sequences create consistent patterns. Sudden changes in these metrics often signal equipment faults, configuration errors, or unintended system modifications—making rate and frequency indicators highly effective for anomaly detection.

Baselines and Thresholds



Baseline

The established pattern representing normal behavior under stable operational conditions



Threshold

The boundary limit beyond which behavior is classified as abnormal and requiring attention



Alert

Notification triggered when measured values exceed the defined threshold



Tuning

The process of adjusting thresholds to reduce false alarms while maintaining detection sensitivity



Drift

Gradual, slow change in baseline behavior over extended time periods



Critical Maintenance Requirement

Baselines must be reviewed and updated when operational conditions change: system upgrades, production schedule modifications, load variations, seasonal patterns, or equipment replacements. Outdated baselines generate excessive false positives and reduce operator confidence.

Alert Outcomes: Four Cases

Every monitoring and detection system produces one of four possible outcomes for each measurement or observation. Understanding these outcomes is essential for evaluating system performance.

True Positive

Correct alert

Real issue exists + Alert raised

System correctly identifies actual problem

False Positive

False alarm

No issue exists + Alert raised

System alerts on normal behavior

False Negative

Missed issue

Real issue exists + No alert

System fails to detect actual problem

True Negative

Correctly quiet

No issue exists + No alert

System correctly remains silent

OT-Specific Risks

Too many false positives → Operators develop **alarm fatigue**, ignore legitimate alerts, and lose trust in the monitoring system

False negatives → Create dangerous **blind spots** that introduce safety and availability risks

False Positives vs False Negatives

False Positive

Creates Alarm Fatigue



Scenario Example: Monitoring system generates alerts during scheduled maintenance windows, even though the behavior is expected and documented.

Impact:

- Operators become desensitized to alerts
- Important warnings may be dismissed
- Time wasted investigating non-issues
- Reduced confidence in monitoring tools

False Negative

Creates Blind Spots



Scenario Example: Gradual pressure buildup goes undetected because it occurs slowly enough to stay below threshold limits.

Impact:

- Real problems go unnoticed
- Safety incidents may occur
- Equipment damage progresses
- Extended downtime and costly repairs

❏ **OT Priority Consideration:** Operational technology environments often prefer **reducing false negatives** for safety-critical signals, even if that means accepting some increase in false positives. Missing a genuine safety issue carries far greater consequences than investigating a few unnecessary alerts.

Precision & Recall

Two fundamental metrics help evaluate the quality and effectiveness of anomaly detection systems. These metrics measure different aspects of detection performance and often involve trade-offs.

Precision

Question answered: "When we raise an alert, how often are we correct?"

High precision means: Fewer false alarms, greater operator trust, less time wasted on investigation

Low precision indicates: Alert noise, operator fatigue, reduced system credibility

Recall

Question answered: "Of all the real issues that occurred, how many did we successfully detect?"

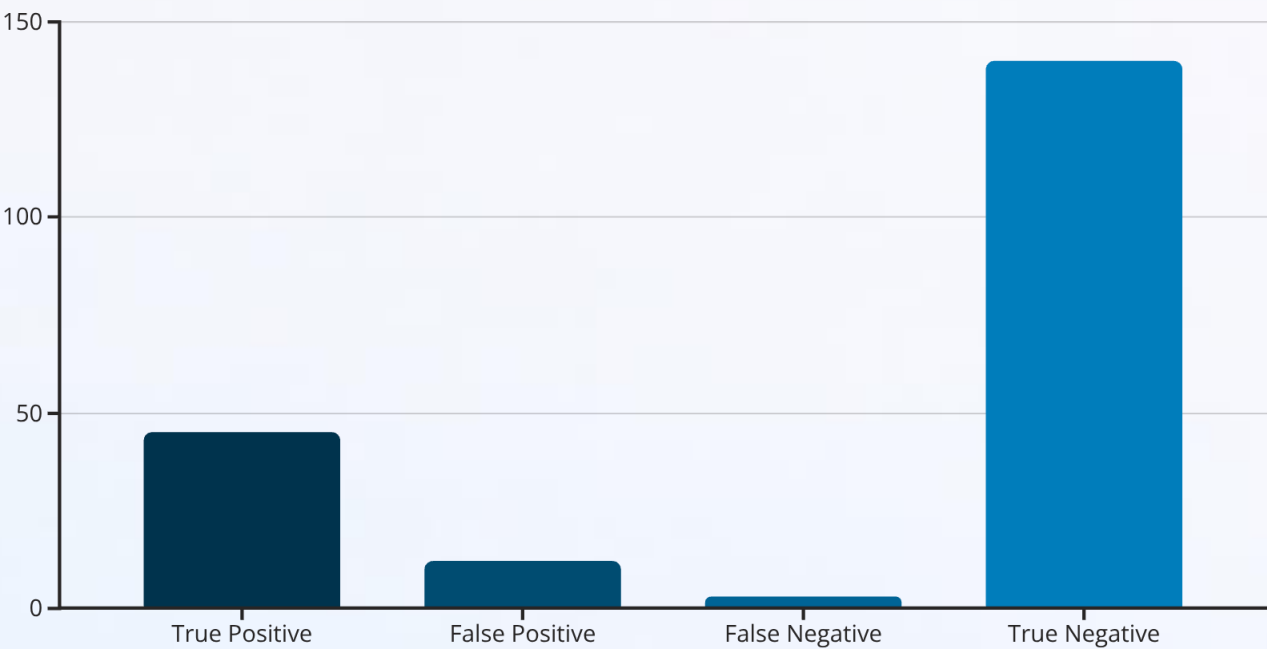
High recall means: Fewer missed incidents, better coverage, improved safety posture

Low recall indicates: Dangerous blind spots, missed problems, incomplete monitoring

Safety-Critical Principle: Safety signals and critical process monitors often require **higher recall**—meaning we prioritize catching real problems even if it means investigating more alerts. Missing a safety event is unacceptable.

Confusion Matrix for Easy Reading

A confusion matrix provides a complete view of detection system performance by summarizing all four possible outcomes. This tool helps identify whether tuning efforts should focus on reducing noise or reducing missed detections.



1	2
Alert Raised + Actual Issue	Alert Raised + No Issue
True Positive — System working correctly	False Positive — Unnecessary alarm, tune threshold
3	4
No Alert + Actual Issue	No Alert + No Issue
False Negative — Dangerous miss, improve sensitivity	True Negative — System correctly silent

Why This Helps: The confusion matrix reveals patterns that guide tuning decisions. High false positive rates suggest overly sensitive thresholds. High false negative rates indicate insufficient detection coverage or poorly chosen features.

Practical Limits of Detection in OT

While anomaly detection provides valuable capabilities, it operates within important practical constraints. Understanding these limitations prevents over-reliance and helps set appropriate expectations.

Data Quality Dependency

Detection effectiveness depends entirely on the **quality and completeness** of available data. Missing sensors, unreliable measurements, or incomplete logging create blind spots that no algorithm can overcome.

Sensitivity to Change

System performance degrades when processes change, schedules shift, or equipment is modified. Detection models trained on historical data may not reflect current operational reality.

Rare Event Challenge

Unusual events that occur infrequently provide insufficient data for learning patterns. Detection systems struggle with scenarios they have never encountered before.

Human Interpretation Required

Alerts and anomalies require **human judgment** for proper interpretation. Context, operational knowledge, and engineering expertise remain essential.

Not for Automated Control

Detection systems should **never** be configured to automatically control processes, shut down equipment, or modify operations. They are decision-support tools only.

❏ **Fundamental Principle:** Detection must remain decision-support, not an automated controller. Human oversight ensures safety and prevents inappropriate automated responses.

Safe Interpretation Habits

When an anomaly alert appears, trained OT personnel should follow a disciplined interpretation process that prioritizes safety, gathers evidence, and maintains operational stability.

- **Check for Known Activities**

Is this deviation linked to scheduled maintenance, planned configuration changes, or documented operational adjustments?

- **Corroborate with Logs**

Do system logs, event records, or operator notes support and explain the observed deviation?

- **Assess Pattern Consistency**

Is the deviation a consistent pattern or a one-time anomaly? Does it repeat across multiple measurements?

- **Confirm with Operations**

Can we safely verify the situation with operations staff without disrupting critical processes?

- **Document Evidence**

What evidence should be captured and documented for review, trending analysis, or incident investigation?

Safe Response Philosophy: Interpret first, act safely, and document clearly. Never rush to judgment or take disruptive action based solely on automated alerts.

Module 09 Key Takeaways

Detection Purpose

OT anomaly detection targets **stability, safety, and availability**—not just cybersecurity threat hunting

Data Foundations

Leverage **events, logs, and time-series data** as complementary inputs for comprehensive monitoring

Simple Features

Prefer **simple, explainable features** such as rates, counts, and frequency metrics that operators can validate

Baselines Matter

Establish clear **baselines** and set appropriate **thresholds** that reflect normal operational behavior

Trade-off Management

Balance false positives (alarm fatigue) against false negatives (missed risk) based on operational priorities

Evaluation Metrics

Use **precision, recall, and confusion matrices** to evaluate and improve alert quality systematically

Human Oversight

Always keep humans in the decision loop—detection supports judgment, it does not replace it

Reflection & Class Discussion

Discussion Topics

- Why anomaly detection in OT focuses on safety and stability, not attacks
- Difference between normal variation and true anomalies
- Why OT anomaly detection should support humans, not replace them
- Impact of false alerts on operators and safety
- Importance of baselines and thresholds in predictable OT systems

Homework & Assignments

1. Explain why anomaly detection is suitable for OT environments
2. List three OT data types used for anomaly detection with examples
3. Describe two simple features extracted from OT logs
4. Differentiate between false positives and false negatives
5. Explain precision and recall using simple OT examples

Applied Learning Task (Optional)

Baseline Exercise:

For a sample OT system (PLC–HMI network):

1. Identify one normal behavior pattern
2. Define a baseline and threshold
3. Describe what condition would trigger an alert

→ COMING NEXT

What's Next: Module 10 Preview

Incident Response and Safety in OT Environments

The next module shifts focus from *detecting anomalies* to **responding safely and restoring operations** when incidents occur. You'll learn the critical differences between OT and IT incident response, and how to maintain safety throughout the response process.



OT vs IT Incidents

Understanding fundamental differences in incident characteristics and priorities



Safety-First Philosophy

Applying safety-first response principles that protect people, equipment, and processes



Roles & Responsibilities

Defining clear incident response roles across operations, engineering, and security teams



Evidence Collection

Documenting incidents properly while maintaining operational continuity



Safe Containment

Containing incidents without disrupting critical processes or creating safety hazards



Recovery & Restoration

Systematic approaches to backup, recovery, and returning to normal operations



Lessons Learned

Capturing insights and improving response capabilities after incidents resolve